1. **What is Data science? Explain the key phases of Data science life cycle process**

Data science is defined as the field of study of various scientific methods, algorithms, tools, and processes that extract useful insights from a vast amount of data.

**Key Phases of the Data Analytics Lifecycle:**

**1. Define Business Objectives/Problem Definition:** Identify the core question or business problem, establish the project scope, and determine key performance indicators (KPIs).

**2. Data Collection/Acquisition:** Identify and gather data from various sources (databases, surveys, APIs) required to answer the business problem.

**3. Data Cleaning and Preparation (Data Wrangling):** Clean the data by removing errors, handling missing values, and restructuring datasets into a usable format (e.g., removing duplicates, fixing data types).

**4. Exploratory Data Analysis (EDA):** Use descriptive statistics and visualization techniques to understand data patterns, trends, and outliers, and to refine the approach.

**5. Data Modeling and Analysis:** Apply analytical techniques, algorithms, or machine learning models to the prepared data to uncover deeper insights and make predictions.

**6. Data Visualization and Reporting:** Present findings to stakeholders using reports, dashboards, or presentations to facilitate data-driven decision-making.

**7. Deployment and Monitoring (Optional/Advanced):** Deploy the model into a production environment and monitor its performance to ensure ongoing accuracy.

2. **Explain any two operations in detail for data preprocessing with examples.**

Data preprocessing is the vital first step in converting messy, raw data into a clean format suitable for analysis or machine learning.
Two critical operations in this process are **Data Cleaning** and **Data Transformation**.

**1. Data Cleaning**

This operation identifies and corrects errors, inconsistencies, and missing information to ensure the dataset is accurate and reliable.

- **Handling Missing Values:** You can address gaps by either removing the affected rows (ignoring the tuple) or filling them using **imputation**.
  - *Example:* If a customer dataset lacks "Age" for some entries, you can use the Scikit-learn Simple Imputer to fill those blanks with the **mean** age of all other customers.
- **Noisy Data Smoothing:** "Noise" refers to random errors or outliers that deviate from the expected trend.
  - *Example:* In a series of temperature readings like [22, 21, 85, 23], the 85 is likely noise. You can use **Binning**—dividing data into segments and replacing values with the bin mean—to smooth these spikes.
- **Removing Duplicates:** Identifying and deleting repeated records to prevent bias in model training.

**2. Data Transformation**

This operation converts data into a specific format or scale that algorithms can process more efficiently.

- **Normalization (Min-Max Scaling):** Rescaling numerical features into a fixed range, typically **0 to 1**.
  - *Example:* If one feature is "Salary" (range $30k–$200k) and another is "Years of Experience" (range 0–40), the model might wrongly prioritize salary due to its larger numbers. Normalizing both to a 0–1 scale ensures they contribute equally.

**Categorical Encoding:** Since many models only understand numbers, categorical text must be converted to numeric codes.

  - *Example:* For a "Color" feature with values like "Red" or "Blue," you can use **One-Hot Encoding** to create separate columns for each color, marked with a 1 or 0.

- **Discretization:** Converting continuous numerical values into discrete "bins" or categories.
- *Example:* Transforming raw ages (e.g., 5, 25, 70) into categories like "Child," "Adult," and "Senior".
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3. **Explain the four types of measurement scales in statistics. Provide examples for each scale and discuss the type of data that each scale is best suited for.**

**Types of Measurement Scale**

**Nominal Scale**

- Definition:

  The nominal scale is used for labelling variables without any quantitative value.

  It is considered a qualitative scale where the labels are mutually exclusive and do not carry numerical significance.

- Characteristics:
  - Labels variables with no inherent order or ranking.
  - No arithmetic calculations can be made.
  - The primary use is for categorization and identification.
- Examples:
  - Gender: Male, Female, Third gender, Other.
  - Languages spoken in a country.
  - Biological species: Human, Dog, Cat.
  - Taxonomic ranks: Archea, Bacteria, Eukarya.
  - Parts of speech in grammar: Noun, Verb, Adjective.
- Applications:
  - Frequency: Rate at which a label occurs in the dataset.
  - Proportion: Frequency divided by the total number of events.
  - Visualization: Can be visualized using pie charts or bar charts.

**2. Ordinal Scale**

- Definition:

  The ordinal scale is similar to the nominal scale but with an additional feature: the order or ranking of the values is significant.

- Characteristics:

  ○ The values can be ordered or ranked.

  ○ The exact difference between the values is not defined.

  ○ Central tendency can be measured using the median; however, mean is not applicable.

- Examples:

  ○ Likert scale (commonly used in surveys): Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree.

- Applications:

  ○ Used to understand <u>the order or rank of categories</u>, but without knowing the exact difference between them.

  ○ Visualization: Bar charts can represent ordinal data.

**3. <u>Interval Scale</u>**

- <u>Definition:</u>

  Interval scales <u>not only have an order</u> but also define the exact difference between values. However, there is no true zero in the interval scale.

- Characteristics:

  ○ The order and exact difference between values are significant.

  ○ Arithmetic operations like mean, median, mode, and standard deviation are applicable.

- Examples:

  - ○ Temperature measured in Celsius or Fahrenheit (e.g., 20°C to 30°C).
  - ○ Dates on a calendar.
- ● Applications:
  - ○ The interval scale allows for a variety of mathematical operations.
  - ○ Visualization: <u>Histograms and line charts</u> can effectively represent interval data.

4. <u>Ratio Scale</u>

● Definition:

The ratio scale includes order, exact values, and an absolute zero point, which enables meaningful ratios to be computed.

● Characteristics:

- ○ It has all the features of an interval scale, plus an absolute zero.
- ○ Can perform all arithmetic operations.
- ○ The measure of central tendencies (mean, median, mode), dispersion, and coefficients of variation can be computed.

● Examples:

- ○ Energy, mass, length, duration, volume.
- ○ Speed, weight, temperature (Kelvin scale).

● Applications:

- ○ Widely used in descriptive and inferential statistics.
- ○ Visualization: <u>Histograms and scatter plots</u> are often used.

Summary of Measurement Scales

| Scale Type | Order | Exact Differences | True Zero | Examples |
|---|---|---|---|---|
| **Nominal** | No | No | No | Gender, Languages |
| **Ordinal** | Yes | No | No | Likert scale |
| **Interval** | Yes | Yes | No | Temperature (Celsius) |
| **Ratio** | Yes | Yes | Yes | Weight, Length, Energy |

4. **Explain different measures of Central Tendency and measures of variability with examples**

Measures of **central tendency** (Mean, Median, and Mode) identify the central point of a data set, while measures of **variability** (Range, Variance, and Standard Deviation) describe how spread out the data points are from that center.

**Measures of Central Tendency**

Measures of central tendency provide a single value that represents the center of a data distribution.

**Mean:** The arithmetic average calculated by summing all values and dividing by the total number of observations ($n$). It is sensitive to outliers.

$$\bar{x} = \frac{\sum x_i}{n}$$

*Example:* For the data set{2,4,6}, the mean is (2+4+6)/3=4

**Median:** The middle value when data is arranged in ascending order. If there is an even number of observations, it is the average of the two middle values. It is robust against outliers.

*Example:* In {3,5,10}, the median is **5**.

**Mode:** The value that appears most frequently in a data set. A set can be unimodal,bimodal,ormultimodal.

*Example:* In {1,2,2,3,4} the mode is **2**.

**Measures of Variability**

Measures of variability (or dispersion) quantify the degree of "spread" or diversity within the data.

- **Range:** The simplest measure, calculated as the difference between the maximum and minimum values.

$$Range = X_{max} - X_{min}$$

*Example:* In $\{10, 20, 50\}$, the range is $50 - 10 = 40$.

- **Variance:** The average of the squared differences from the Mean. It indicates how much the data points fluctuate around the center.

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$$

- **Standard Deviation:** The square root of the variance. It expresses the spread in the same units as the original data, making it the most commonly used measure of dispersion.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

- **Interquartile Range (IQR):** The difference between the third quartile ($Q_3$) and the first quartile ($Q_1$), representing the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

5. **What is Data visualization? List and identify the applicability of different types of plots.**

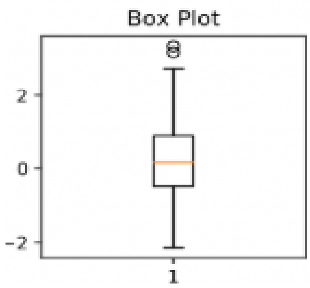
Data visualization is the graphical representation of information and data using visual elements like charts, graphs, and maps, designed to make complex data sets accessible, understandable, and actionable. It facilitates the identification of patterns, trends, and outliers, allowing for faster data-driven decision-making.

**Common Plots and Their Applicability**

The choice of a plot depends on the type of data and the specific analytical goal, such as comparison, distribution, or relationship analysis.

- **Bar Chart**: Best for **comparing quantities** across different discrete categories. Use horizontal bar charts if category names are long or numerous.
- **Line Graph**: Ideal for tracking **trends over time** across continuous measurements. It is standard for financial and website traffic analysis.

- **Pie Chart**: Suitable for showing **proportions of a whole** for a small number of categories (ideally fewer than six).
- **Scatter Plot**: Used to explore **correlations and relationships** between two numerical variables. It is also effective for identifying outliers and data clusters.
- **Histogram**: Designed to show the **distribution and frequency** of continuous numerical data by grouping values into intervals (bins).
- **Box Plot (Box-and-Whisker)**: Essential for **statistical summaries**, displaying the median, quartiles, and variability while highlighting outliers.
- **Heatmap**: Uses color gradients to represent **data intensity** or frequency across a 2D matrix or geographical area.


heat map diagram

6. **Demonstrate the applicability of Naïve Bayes with an example.**

Naïve Bayes is a probabilistic machine learning algorithm based on Bayes' Theorem, widely used for classification tasks like spam filtering, sentiment analysis, and document categorization. It calculates the probability of a class (e.g., Spam/Ham) given input features (e.g., words) assuming feature independence.

**Example: Spam Email Classification**

**Scenario:** Determine if an email containing the words "Winner" and "Cash" is Spam or Not Spam.

**1. Data Preparation (Training Phase):**

- **Total Emails:** 10
- **Spam Emails:** 4
- **Ham (Not Spam) Emails:** 6
- **Word "Winner" in Spam:** 3 times
- **Word "Winner" in Ham:** 1 time
- **Word "Cash" in Spam:** 3 times
- **Word "Cash" in Ham:** 2 times

**2. Probabilities Calculation:**

- **Prior Probability:** $\boldsymbol{P(\text{Spam}) = \frac{4}{10} = 0.4}$
- **Likelihoods:**

$$\boldsymbol{P(\text{Winner}|\text{Spam}) = \frac{3}{4} = 0.75}$$

**Assumption of Independence:**

$$P(\text{Winner} \cap \text{Cash}|\text{Spam}) = P(\text{Winner}|\text{Spam}) \times P(\text{Cash}|\text{Spam}) = 0.75 \times 0.75 = 0.5625$$

$P$(Winner∩Cash|Spam)=$P$(Winner|Spam)×$P$(Cash|Spam)=0.75×0.75=0.5625

**3. Applying Bayes' Theorem for a new email ("Winner" + "Cash"):**

- **Posterior Probability (Spam):**

$$\propto P(\text{Spam}) \times P(\text{Winner}|\text{Spam}) \times P(\text{Cash}|\text{Spam})$$

$$0.4 \times 0.75 \times 0.75 = 0.225$$

- **Posterior Probability (Ham):**

$$\propto P(\text{Ham}) \times P(\text{Winner}|\text{Ham}) \times P(\text{Cash}|\text{Ham})$$

$$0.6 \times \frac{1}{6} \times \frac{2}{6} = 0.6 \times 0.166 \times 0.333 = 0.03$$

**Conclusion:**
Since the probability of it being Spam (0.225) is significantly higher than Ham (0.033), the classifier classifies the email as **Spam**.

**Key Advantages**

- **Fast and Efficient:** Requires small training data and is computationally fast.
- **High Performance:** Works well with multi-class prediction and text categorization.

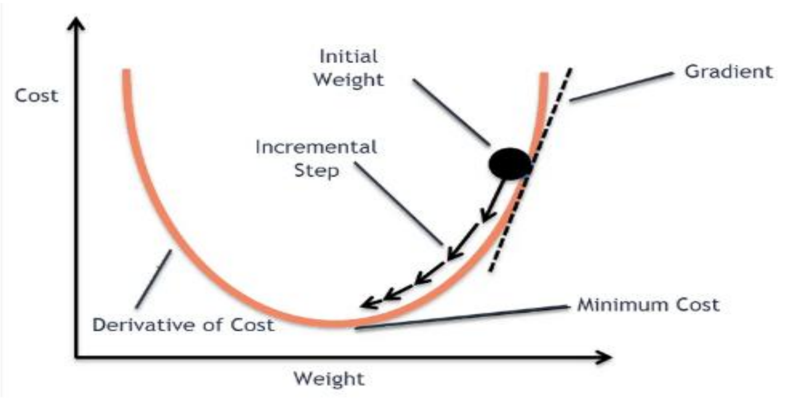
7. **Demonstrate on Gradient Descent optimization and its variants with appropriate diagrams.**

Gradient descent is an iterative optimization algorithm used to minimize a cost function by moving in the direction of the steepest descent. It is a foundational tool for training machine learning models like neural networks and linear regression

**Core Optimization Mechanism**
The algorithm starts with random parameter values and iteratively updates them.
The update rule is:

$$\theta = \theta - \eta \cdot \nabla_\theta J(\theta)$$

$\boldsymbol{\theta}$: Model parameters (weights).
$\boldsymbol{\eta}$: Learning rate, which controls the step size.
$\boldsymbol{\nabla\theta J(\theta)}$: The gradient (slope) of the cost function $J$



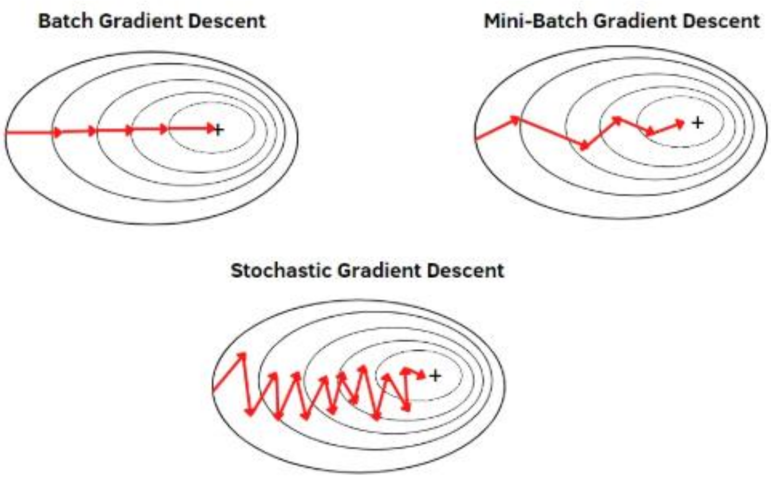

**Variants of Gradient Descent**

Variants differ primarily in the amount of data used to calculate the gradient in each step.

**Batch Gradient Descent (BGD):** Computes the gradient using the **entire dataset** for every update. It provides a stable, smooth path toward the minimum but can be very slow and memory-intensive for large datasets.

**Stochastic Gradient Descent (SGD):** Updates parameters based on a **single data point** at a time. This is much faster and helps the model escape local minima due to its high "noise" or randomness, though it never truly settles at the exact minimum.

**Mini-Batch Gradient Descent:** A compromise that updates parameters using **small subsets (batches)** of data. It is the standard choice for deep learning as it balances computational efficiency with stable convergence




**Advanced Adaptive Variants**

Beyond data-usage variants, modern optimizers adjust the **learning rate** dynamically:

**Momentum:** Adds a fraction of the previous update to the current one to "speed up" through flat regions and reduce oscillations.

**RMSprop:** Adjusts the learning rate for each parameter by using a moving average of squared gradients.

**Adam (Adaptive Moment Estimation):** Combines Momentum and RMSprop, making it one of the most effective optimizers for complex neural networks.

## 8. Demonstrate on Linear regression model with a neat diagram.

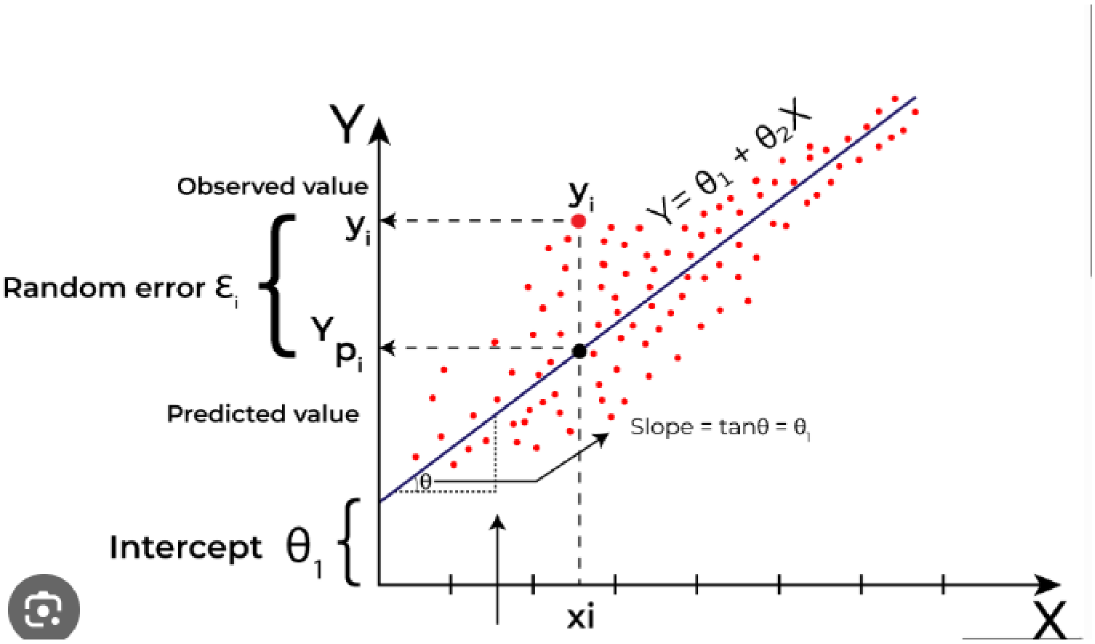
Linear regression is a statistical method used to model the relationship between a **dependent variable** ($Y$) and one or more **independent variables** ($X$). It finds the "best-fitting" straight line that minimizes the distance between the observed data points and the line itself.

### The Linear Regression Model Equation

For simple linear regression, the relationship is expressed as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- $Y$: The dependent variable (e.g., house price).
- $X$: The independent variable (e.g., house size).
- $\beta_0$: The **Y-intercept**, where the line crosses the vertical axis.
- $\beta_1$: The **slope**, indicating how much $Y$ changes for every unit increase in $X$.
- $\epsilon$: The **error term** (residual), representing the difference between actual and predicted values.




For example we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied. Here

- Independent variable (input): Hours studied because it's the factor we control or observe.
- Dependent variable (output): Exam score because it depends on how many hours were studied.
  More explanation can be included

**9. Demonstrate the applicability of boxplot with a neat diagram.**

**10. Explain different types of probability distribution functions with neat diagrams.**

**11. Explain the concept of Bayes theorem with an example.**

**12. What is EDA?explain its significance and steps involved**

Exploratory Data Analysis (EDA):

EDA is the process of analyzing data sets to summarize their main characteristics, often with visual methods.

It is a critical initial step in any data mining project and plays a key role in making sense of large datasets.

**Steps in EDA:**

The process of Exploratory Data Analysis (EDA) involves four main steps that guide the data analysis from problem definition to presenting results.

These steps are critical for structuring the analysis, preparing the data, and interpreting the results effectively.

1. Problem Definition:

● Importance: The first step in any data analysis is defining the business problem to be solved.

The problem definition serves as the foundation and guides the entire analysis process.

● Key Tasks:

○ Defining the Main Objective: Clarifying the problem to be solved with the data.

○ Defining Deliverables: Identifying what outputs or results will be expected from the analysis.

○ Outlining Roles and Responsibilities: Assigning tasks to different team members or stakeholders.

○ Obtaining Current Data Status: Assessing the current state of the data before analysis begins.

○ Defining the Timetable: Setting a timeline for completing the analysis.

○ Cost/Benefit Analysis: Evaluating the financial and resource implications of conducting the analysis.

2. Data Preparation:

● Importance: Data preparation is critical to ensure that the dataset is clean, well structured, and ready for analysis.

● Key Tasks:

○ Defining Data Sources: Identifying where the data will come from.

○ Defining Data Schemas and Tables: Structuring the data and setting up the necessary tables.

○ Understanding Data Characteristics: Analysing the data for its key features.

○ Data Cleaning: Handling missing or erroneous data points.

○ Removing Non-Relevant Datasets: Eliminating data that is not useful for the analysis.

○ Data Transformation: Converting the data into a format that can be used for analysis.

○ Dividing Data into Chunks: Breaking the data into smaller, manageable parts.

3. Data Analysis:

● Importance: This step involves applying statistical techniques to the dataset to derive meaningful insights.

● Key Tasks:

○ Summarizing Data:Using summary statistics (mean, median, mode, etc.) to describe the data.

○ Finding Correlations and Relationships: Identifying any connections between variables.

○ Developing Predictive Models: Creating models to predict future outcomes or trends.

○ Evaluating Models: Assessing the performance and accuracy of predictive models.

○ Calculating Accuracies: Using various metrics (e.g., precision, recall) to evaluate model accuracy.

● Techniques Used:

○ Summary Tables, Graphs

○ Descriptive and Inferential Statistics

○ Correlation Statistics

○ Mathematical Models

4. Development and Representation of Results:

● Importance: After analysis, it is crucial to represent the findings in a clear and interpretable format for stakeholders.

● Key Tasks:

○ Graphical Representation: Presenting results using charts and plots.

○ Summary Tables and Maps: Using tables to summarize key findings.

○ Communicating Results Effectively: Ensuring that business stakeholders can easily understand the results of the analysis.

● Graphical Analysis Techniques:

○ Scatter Plots

○ Character Plots

○ Histograms

○ Box Plots

○ Residual Plots

○ Mean Plot

13. Differntiate between

a) Qualitative approach and Quantitative approach

b) Structured and Unstructured Data

14.